Fungal species richness in logs at different stages of decay

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EAP Tropical Biology and Conservation Spring 2025
6 June 2025

ABSTRACT.

This study examines species richness and diversity of fungi on decaying logs and its intersection with five different decay stages and presence of other fungal species on same logs in order to find more evidence for fungal succession. This was completed through measuring decaying logs with a visible presence of fruiting bodies found on hiking trails in Monteverde, Costa Rica. These measurements included average depth from knife penetration (quantifies decay stage) and log volume measurements. In order to identify species or genera, fruiting bodies were observed and noted on texture, color, and shape, most of which were photographed. Out of the fifty one logs, eighty four individual fungal populations were found, including fifty six species, genera, or families that were identified. Results showed that diversity has an inverse relationship with log size and is highest in weak decay stages and lowest in almost decomposed stages, both of which contradicting previous studies. *Chlorociboria aeruginascens* showed the best supporting signs of being a pioneer species and potential predecessor-successor relations with other species. This was due to *C. aeruginascens* almost exclusively being found on logs by itself. Whereas the other two abundant genera, *Ductifera sp.* and *Xylaria spp.*, were found frequently on logs with other species, but due to decay data, showed only little conclusive evidence of having a role in fungal succession.

Riqueza de especies fúngicas en troncos en diferentes etapas de descomposición

RESUMEN

Este estudio examina la riqueza y diversidad de especies de hongos en troncos en descomposición y su relación con cinco etapas distintas de descomposición y la presencia de otras especies fúngicas en los mismos troncos, con el objetivo de encontrar más evidencia sobre la sucesión fúngica. La investigación se llevó a cabo mediante el análisis de troncos en descomposición con presencia visible de cuerpos fructíferos, encontrados en senderos de Monteverde, Costa Rica. Las mediciones incluyeron la profundidad promedio de penetración de un cuchillo (para cuantificar el grado de descomposición) y el volumen del tronco. Para identificar especies o géneros, se observaron los cuerpos fructíferos y se documentaron características como textura, color y forma; la mayoría fueron fotografiados. De los cincuenta y un troncos analizados, se encontraron ochenta y cuatro poblaciones fúngicas individuales, incluyendo cincuenta y seis especies, géneros o familias identificadas. Los resultados mostraron que la diversidad tiene una relación inversa con el tamaño del tronco y es mayor en etapas de descomposición débil y menor en etapas casi completamente descompuestas, lo cual contradice estudios previos. Chlorociboria aeruginascens mostró signos consistentes de ser una especie pionera, así como posibles relaciones de precedencia con otras especies, debido a que se encontró casi exclusivamente en troncos donde no había otras especies presentes. En contraste, los otros dos géneros abundantes, Ductifera sp. y Xylaria spp., fueron encontrados frecuentemente en troncos con otras especies, pero los datos de descomposición mostraron poca evidencia concluyente de su participación en procesos de sucesión fúngica.

INTRODUCTION

Mycology has since been an understudied field in science. However, its information and breakthroughs in this field are just as important as those in other fields. Fungi in general have been proposed as holders of a significant role in regulating energy and nutrient production and cycling in natural ecosystems (Dighton, 2007). This applies to a wide range of fungi including mycorrhizae and saprotrophs. These fungi's nutrient and energy cycling abilities are understood to a basic level, however, certain cause and effect relationships are commonly questioned. This includes saprotrophic and mycorrhizal fungi and their cycling productivity during periods of unstable climate conditions (Dighton, 2007). This curiosity in particular holds a great amount of importance, especially given how the earth has progressed during the rapid climate change.

Scientists have found various suggested patterns between saprotrophic fungi and certain wood log criteria and qualities. In one study, there was evidence to suggest a strong relationship log size and fungal diversity. Specifically, a pattern was determined in which fungal diversity increased with also increasing log volume in New Zealand (Allen et al, 2000). Other qualities were tested in the same study as well, including nitrogen, phosphorus, potassium, calcium, and magnesium properties within the logs. However, the study could only determine a weak relation between the given properties and fungal diversity, suggesting that such properties are not determinants in fungal diversity (Allen et al, 2000).

Diversity in this field has also been explored in terms of decay stages of logs. By looking at species richness and identification of present species at different stages of decay in wood logs, the evidence could suggest succession patterns. In one study, it was found that highest species diversity occurred in intermediate decay logs, with minimal decay showing the least amount of diversity (Farinacci, 2004). Furthermore, certain species were found to be more abundant at one stage of decay rather than another, supporting the idea of fungal succession (Farinacci, 2004). Distribution was also studied in terms of fungal communities and decaying logs.

Specifically, distance between decaying logs has been explored by the scientific community. In Norway, there was no evidence supporting a slight correlation between the species found and log distance from each other (Kubartoá et al, 2012). It was found that log distance between each other did not have a significant correlation to the similarity between fungal communities (Kubartoá et al, 2012). However, distance between communities on the same log had a slight correlation in resemblance in genetic coding to each other (Kubartoá et al, 2012).

General fungal diversity has been studied and accounted for at different decay stages, log volume, nutrient properties, and distance between and within decaying logs. However, species within the same logs have not been studied much. The closest study to such a topic included community resemblance in same logs, but the study itself was not primarily focused on the individual logs, but rather the whole area of logs (Kubartoá et. al, 2012). Specifically, diversity in fungal species on individual logs and any such correlations specifically between the species and on log's decay stages, is understudied, if at all. By examining any logs showing signs of a present fungal population/community and quantifying decay stages, I aim to answer if fungal species richness is impacted by specific decay stages and by the presence of other species on the same log.

MATERIALS AND METHODS

In order to carry out this study, I looked for decaying logs with signs of a present fungal population (i.e. visible fruiting bodies). I included a log in the sample if I saw mushrooms or fruiting bodies growing from said log. This study took place in Sendero Principál at the Estación Biológica, the Crandall Reserve at the Monteverde Institute, and the Guindon Farm, all in Monteverde, Costa Rica. Samples were collected on eleven different days from 12 May 2025 to 25 May 2025. Four days of collection were spent on Sendero Principal and one day at the Guindon Farm. The six other collection days were spent at the Crandall Reserve. On the days that I was not collecting log samples, time was being spent carefully identifying the fungal organisms found present. Furthermore, any logs that I found with fruiting bodies within three meters of these trails were eligible to be a part of the whole log sample.

I started by measuring the length and diameter of each log, with the assumption that each log has a perfect circle structure. These measurements were made with a one-meter long measuring tape. From these measurements, I calculated the rough volume of each log. This is because I was looking to determine which size category to put the log in, determined by specific ranges. These ranges were 15,999 cm^3 and less for small logs, 16,000-80,999 cm^3 for medium logs, and 81,000 cm^3 and above to be considered large logs. Then, with a butter knife, I pierced the log in three different places, once on each end and another in the middle. I would measure in length how much of the knife had been fully pierced into the log. From this measurement, I am looking to determine the log's current stage of decay with specific depth ranges that the knife pierces. As per a previous study, I divided the logs into five stages of decay (Maniken et al., 2006). The first is a recently dead tree which ranges from 0-1 cm of depth (Maniken et al., 2006). Next, any depths recorded above 1 cm and below 2 cm represent weakly decayed logs, while medium decay is represented by 2-5 cm of knife penetration (Maniken et al., 2006). Very decayed logs and almost decomposed logs are very similar because they have the same numerical range, which is anything above 5 cm (Maniken et al., 2006). However, an almost decomposed log will be overgrown with lichens, mosses, and other species aside from fungi, and very decayed logs will strictly have fruiting bodies (Maniken et al., 2006). Once the knife had been stabbed, I kept it in the log and, marked with a pen, right above the bark line to determine how far the knife went into the log. I then traced this marking and knife in my field notebook and later determined the length with the measuring tape. For this measurement, I assumed that my pressure and strength of stab were the same each time, and did not take into account the log's species. Lastly, I took pictures of all of the fruiting bodies present, as well as physically examined the mushroom with my hands to determine its texture and structure.

All of the characteristics that I determined were written in my field notebook in order to most accurately identify each fruiting body. I have been using various modes of media in order to identify each fruiting body including ID books, iNaturalist, and other highly knowledgeable scientists including Naomi Solano Guindon, Jose Murilla, Alan Rockfeller, and Ryan Mcdowell. Each mushroom has thus far been identified as specifically as possible, preferably to the species name, but if not, I determined the family or at the very least its genus. No morphospecies were identified.

RESULTS

I found 51 decaying logs with fruiting bodies at different decay stages and sizes. Specifically, 25 small, 13 medium, and 13 large logs were found. In terms of decay, most logs found are within the weakly decayed stage, accounting for 23 of the logs in the sample. The medium decayed stage had the next highest number of logs in the sample with 11. There were 9 logs found in the first decay stage, recently dead trees, 5 very decayed, and 3 almost decomposed. In terms of species count, 56 different species and 84 individual populations of fruiting bodies were found throughout the sample of logs. *Xylaria hypoxylon*, *Ductifera pululahuana*, and *Chlorociboria aeruginascens* being the three most occuring species within the entire sample. *Xylaria hypoxylon* was found on seven separate logs, but mostly on small logs and weakly decayed logs. *Ductifera pululahuana* and *Chlorociboria aeruginascens* were both found on 6 separate logs each. However, *Ductifera pululahuana* was more abundant on medium and large logs while *Chlorociboria aeruginascens* was most abundant on small logs. Both were more abundant on weakly decayed logs as well.

The average number of different species found on each log category, size and decay stage, was found. For small logs, the average was about one species found on each log. This is different from medium and large, both averaging 1.5 species/log. Figure 1 shows these values on a bar graph. For the decay stages, recently dead and weakly decayed had the same rate of 1.2 species/log, with medium decay right below at 1.1 species/log. The highest average number of species found/log was in the very decayed categories at 1.8 species/log, with almost decomposed right below with 1.7 species/log. Figure 2 shows these values on a bar graph in order to visually compare categories.

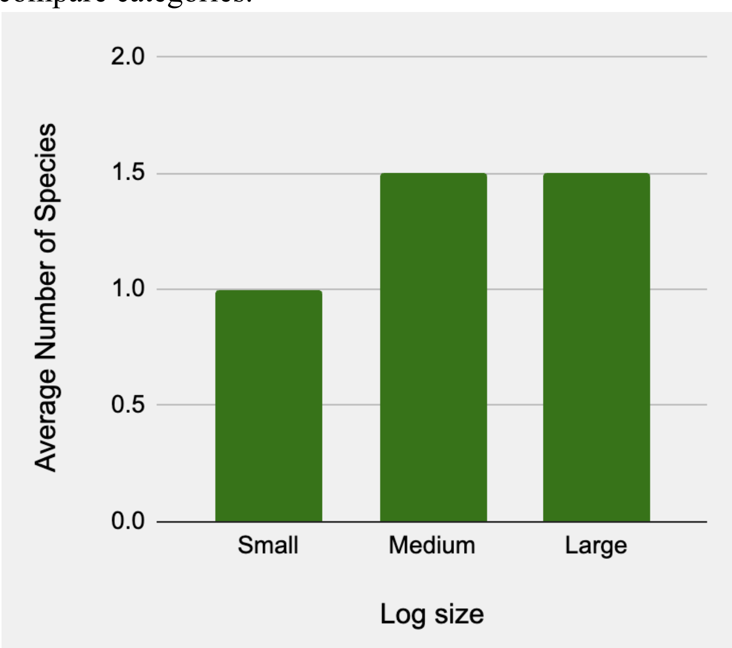

Figure 1. Bar graph showing the average number of different species found on each log size. The X-axis represents the category, i.e. log size, whereas the Y-axis represents the average number of species found. Small logs = 1 species/log, medium logs = 1.5 species/log, and large log = 1.5 species/log on average.

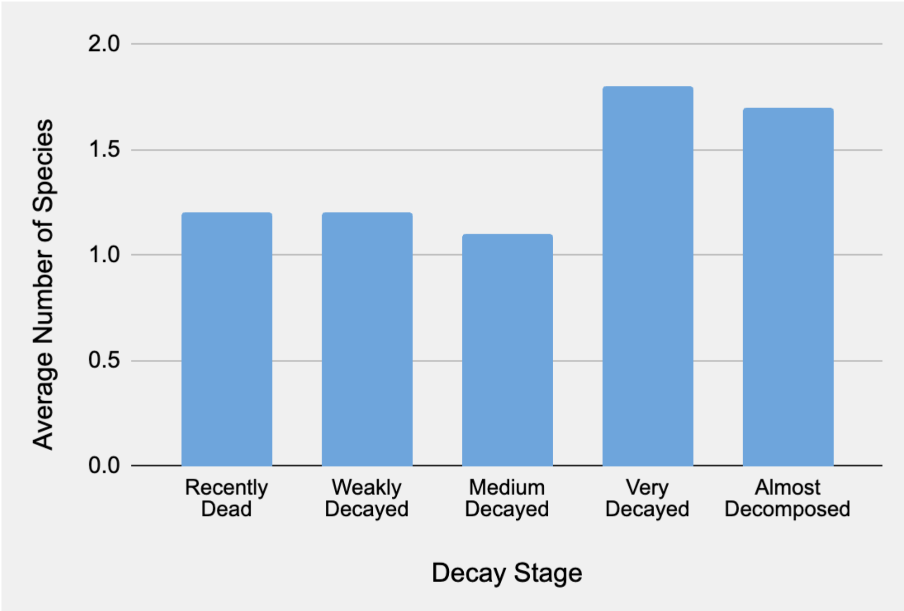

Figure 2. Bar graph showing the average number of different species found on each decay stage of log. The X-axis represents the category, i.e. decay stage, whereas the Y-axis represents the average number of species found. Recently dead trees = 1.2 species/log, weakly decayed = 1.2 species/log, medium decayed = 1.1 species/log, very decayed = 1.8 species/log, and almost decomposed = 1.7 species/log on average.

These next plots are made of the number of populations that each log category was found present with. For small logs, the maximum number of populations found was four with a median of one. The average number of populations for this size was 1.4 ± 0.8. For the medium and large sized logs, the maximum number of populations found was three with a median of two, however these log sizes differ in mean values of populations found. The average number of populations found on medium logs was 1.7 ± 0.6, and 1.8 ± 0.7 for large logs. When looking at the graph (Fig. 1), mean values can be found at the X's of each box and whisker plot. The figure 1 displays the relationship between the number of individual populations found and the size of log they were found to be present on. By looking at the figure, it is clear that minimum, quartile one (Q1), and quartile three (Q2) values are the same for each size (min = 1, Q1 = 1, Q3 = 2), whereas mean, median, and maximum values differ slightly between the three sizes.

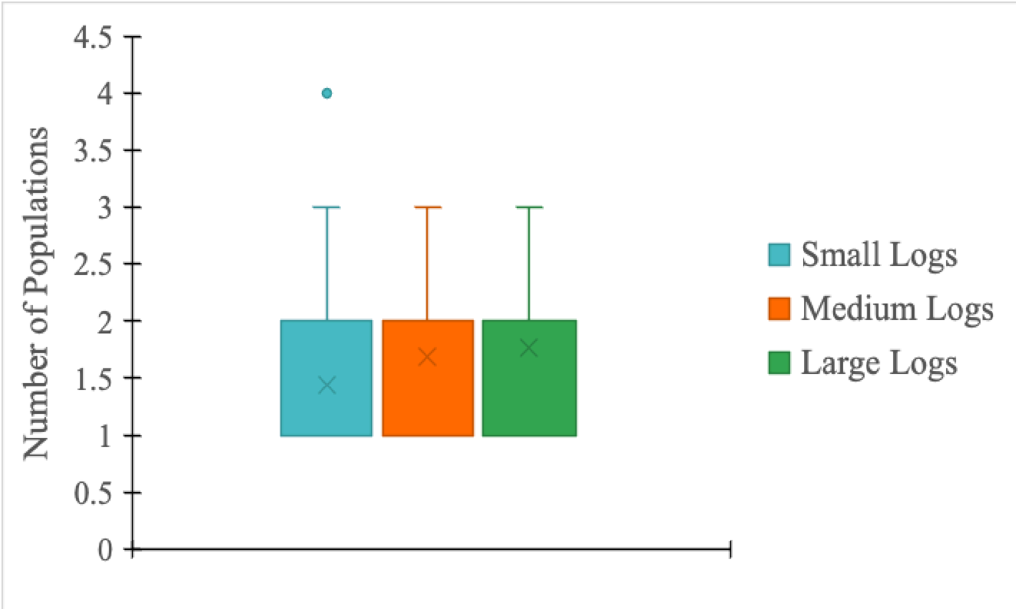

Figure 3. Box and whiskers plot for the number of mushroom populations found on each log size. Specific log size categories (small logs, medium logs, large logs) are color coded as shown in the figure legend. The mean values are as follows, small logs = 1.4 populations, medium logs = 1.7 populations, large logs = 1.8 populations.

The average number of populations found on recently dead trees was 1.4 ± 0.7 and 1.3 ± 0.5 for medium decayed logs. Weakly decayed and very decayed logs also share the same median values (median = 2), but differ slightly in Q3, maximum, and mean values. Weakly decayed logs had a Q3 value of 2, a maximum value of 4, and an average of 1.7 ± 0.9 populations found. Very decayed logs had a Q3 value of 2.5, a maximum value of 3, and an average of 1.8 ± 0.8 populations found. When looking at the graph (Fig. 4), mean values can be found at the X's of each box and whisker plot. The second graph (Fi. 4) depicts the relationship between the number of individual populations found and the decay stage of the log. This was made with the same method as Figure 3, but in terms of the decay stage categories as opposed to size categories. By looking at the figure, it is clear that there are certain major differences in values between a couple decay stages. In particular, the last stage, almost decomposed, has all of its five point summary values at two individual populations, including its mean, which is 2 ± 0. All of the other decay stages have the same minimum and Q1 values, all equaling 1 population found. Recently dead trees and medium decayed trees also have the same median, Q3, and maximum values (median = 1, Q3 = 2, max = 3), while their mean values of populations found differ slightly.

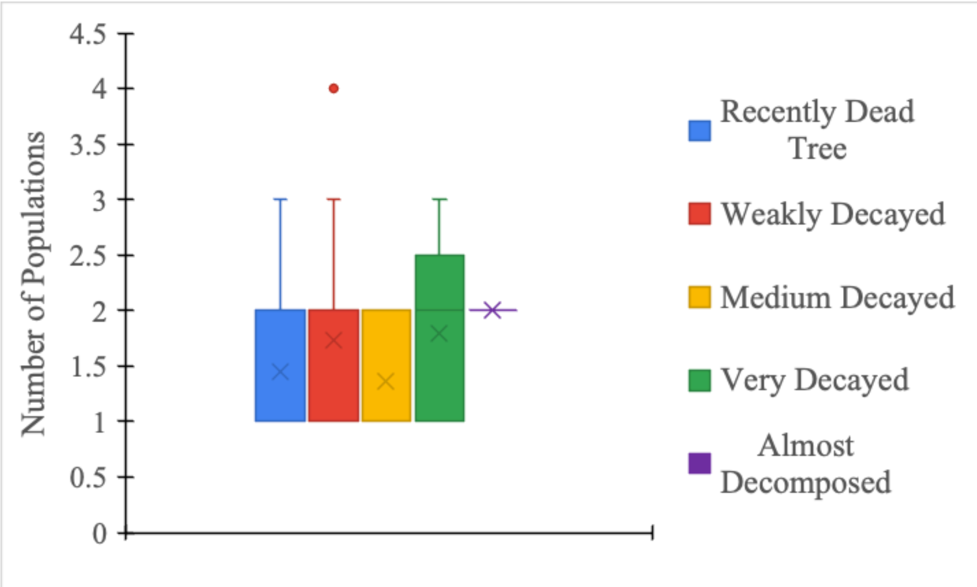

Figure 4. Box and whiskers plot for the number of mushroom populations found on each of the five decay stages of wood logs. Specific decay stage categories (recently dead tree, weakly decayed, medium decayed, very decayed, almost decomposed) are color coded as shown in the figure legend. The mean values are as follows, recently dead tree = 1.4 populations, weakly decayed = 1.7 populations, medium decayed = 1.3 populations, very decayed = 1.8 populations, almost decayed = 2 populations.

Two diversity indexes were calculated in order to understand and find any patterns between fungal species diversity and log size (fig. 5) or decay stage (fig. 6). Shannon and true diversity indexes were calculated. Weakly decayed logs had the highest diversity values with both shannon (H' = 3.13) and true diversity (eH' = 22.94) indexes. Almost decomposed had the lowest diversity with H' = 1.56 and eH' = 4.76. Recently dead (H' = 2.35 and eH' = 10.50), medium decayed (H' = 2.51 and eH' = 12.34), and very decayed (H' = 2.20 and eH' = 9.00) had very similar diversity calculations. Figure 5 shows this relationship in a color coded column bar graph, with the Shannon index (H') in green bars and the true diversity index (eH') in orange bars.

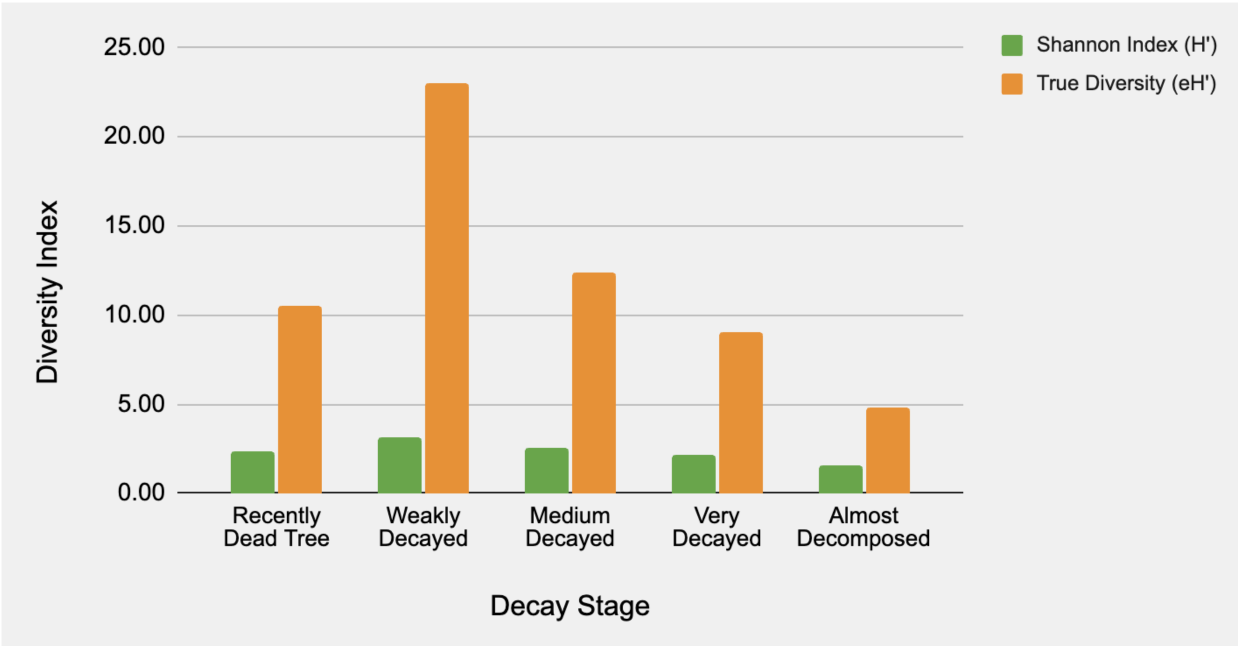

Figure 5. Shannon diversity index and true diversity indexes were calculated and depicted on this bar graph in order to visually compare each value by decay stage. Shannon diversity is shown in the green bars and true diversity is shown in orange. Weakly decayed had the highest diversity in both indexes H' = 3.13 and eH' = 22.94. Medium decayed was next with H' = 2.51 and eH' = 12.34, recently dead tree H' = 2.35 and eH' = 10.50, very decayed H' = 2.20 and eH' = 9.00, and almost decomposed had the lowest diversity with H' = 1.56 and eH' = 4.76.

The second graph (fig. 6)comparing different diversity indexes (Shannon and True Diversity) is shown in figure 6. However, instead of being compared by decay stage, the log sizes were taken into account. Results showed diversity decreased with size increase. Small sized logs with the highest diversity (, then medium sized logs (), and large sized logs (). The bar graph visually compares each of the size category's indices values. It is also color coded with Shannon diversity (H') shown in green and true diversity (eH') shown in orange.

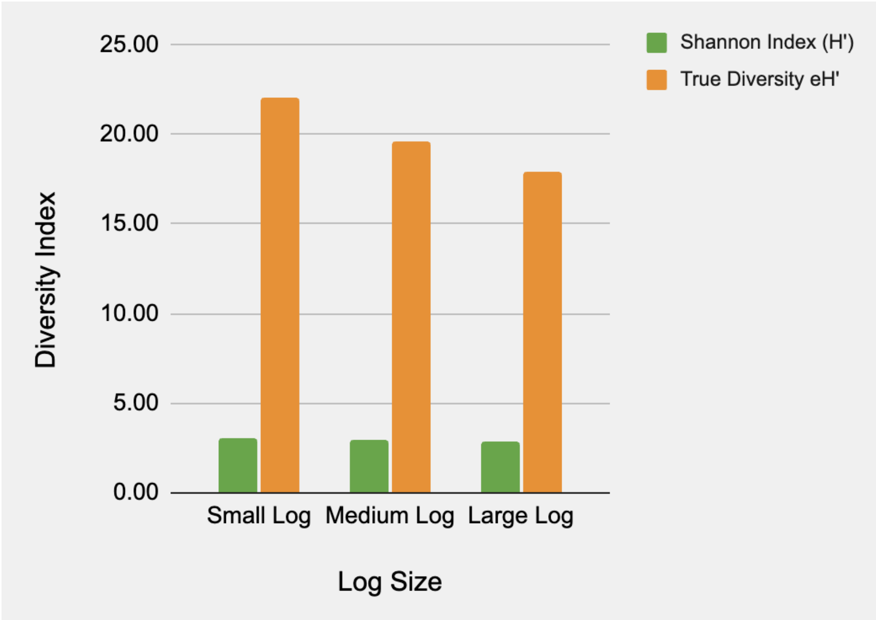

Figure 6. Shannon diversity index and true diversity indexes were calculated and depicted on this bar graph in order to visually compare each value by decay stage. Shannon diversity is shown in the green bars and true diversity is shown in orange. Small sized logs had the highest diversity in both indexes H’ = 3.09 and eH’ = 22.08. Medium sized logs were next with H’ = 2.98 and eH’ = 19.61, and large sized logs with H’ = 2.89 and eH’ = 17.92.

Two tables were also constructed in order to better visualize where each of the 56 species of mushrooms were found present. Table 1 in the appendix shows each of the 56 mushroom species, which size log, and how many times it was found present on each size. At the bottom of the table, totals are calculated for each log size in terms of the number of different species that were found, as well as a total of general species abundance. Most species are found on only one size of log. Only two species were found on all sizes of logs, which included *Ductifera pululahuana* and *Xylaria hypoxylon*. Overall, the small sized logs had the highest number of different species present, with 26 species found and an overall species abundance of 36. Medium sized logs had the next highest values, having 20 different species and an overall species abundance of 25. Last was the large sized logs, with 19 different species and an overall species abundance of 23.

The second table (Table 2 in the appendix) depicts each of the 56 mushroom species found, which decay stage of log, and how many times it was found present on each stage. Similar to Table 1, the bottom of Table 2 includes calculated totals for each decay stage in terms of the number of different species that were found, as well as a total of general species abundance. The weakly decayed stage of logs had the highest number of different species present, with 28 species found and an overall species abundance of 36. Medium decayed logs had the next highest values, with 12 different species and an overall species abundance of 16. Then recently dead trees, with 11 different species found and an overall species abundance of 13. Very decayed and almost decomposed logs had the lowest values, very decayed with both values equaling 9 species, and almost decomposed having 5 species found and an overall species abundance of 6. No species were found on all five stages of decay in logs.

A third table, in the appendix was constructed to visualize and understand the present fungal communities on logs in the sample. This table consists of the 25 logs in which two or more species were present at one time. Species that only occurred on logs by themselves are not included in Table 3. Species *Xylaria hypoxylon* and *Ductifera pululahuana* were found present the most with other fungal species. *X. hypoxylon* was present on logs with other fungal species, and *D. pululahuana* was present in five. *Chlorociboria aeruginascens* was only found on one log containing other fungal species, despite being one of the most abundant species in the sample, along with *X. hypoxylon* and *D. pululahuana*.

DISCUSSION

While looking back at the literature that I had reviewed before beginning my research, there are some key findings that relate to the results that I had received from my study. I found from my results that species diversity was highest at the weakly decayed stage, with medium decayed with the next highest diversity below weakly decayed. This is similar to Farinacci's (2004) findings, that fungal species diversity was highest in intermediate decay stages. However, there was an interesting difference in our results. Farinacci (2004) also found that fungal species diversity was smallest in minimally decayed logs, which was not the case in my study. Lowest species diversity was found in the almost decomposed range, the last stage of decay, which contradicts previous studies (Farinacci, 2004). However, my sample size of logs in the last stage of decay is very disproportionate compared to others. The almost decomposed stage had only three samples, being the smallest sample size, whereas the weakly decayed stage had the largest with 23 log samples. Due to this discrepancy, more research with higher and more even sample sizes is needed to determine a distinct pattern between fungal species diversity and decay stage.

Relating my results of species diversity and log size with my previously reviewed. It was found in previous studies that species diversity increased as the size of the log increased (Allen et al., 2000). This was also not congruent with the results of my study. I found the opposite, in that species diversity had an inverse relationship with log size. As log size increased, the number of species (and individual populations) decreased. However, I believe this discrepancy is also due to disproportions in sample sizes within each category. Small logs were the most abundant in the sample with 25 logs found, whereas large and medium logs had the same, much smaller sample size of 13 logs. Similar to suggested further research for decay stages and diversity in fungal species, the relationship between log size and diversity would be best determined through another study with higher and more even sample sizes.

In order to understand succession in relation to fungal species and decaying wood logs, it is important to understand preferences and tendencies of major species and genera. This is explored in the three most abundant genera in the entire sample, *Chlorociboria aeruginascens*, *Xylaria*, and *Ductifera pululahuana*. Specific tendencies and preferences that are important to note when answering the question of succession include seasonality, the decay stage of the log in which the species or genera is found to be present on, and presence in fungal communities (i.e. found on logs with other species present).

One of the most common species in the sample, *Chlorociboria aeruginascens*, showed interesting tendencies or potential preferences when it came to log size, decay, and accompanying species. Particularly, this species was most often found on small and weakly decayed logs. When not found present with those log characteristics, this species was present on medium sized, recently dead trees. *C. aeruginascens* was also found (five times out of six) on logs all by itself, no other species or genera present. With these results, there is support for the idea that this species is a pioneer species, coming before all others to start the process of decay in a log. However, other successional interactions could be suggested here as well. It was found in other locations that some species have successional preferences, but not determined by log size or decay stage (Boddy & Hiscox, 2016). Species *Antrodiella hoehnelii* was consistently observed fruiting on a log after species *Inonotus nodulosus* and *Inonotus radiatus* had lived out their own life cycle on the log (Boddy & Hiscox, 2016). This heavily suggests the idea of predecessor-successor relationships between saprotrophic fungal species, which could also be studied in species like *Chlorociboria aeruginascens*. Particularly, further research could be done on this topic with *Chlorociboria aeruginascens* by continually observing the logs in which they are found present, taking note and identifying other species that may or may not fruit after *C. aeruginascens* has gone through its life cycle. I believe that it is likely for this species to have some sort of predecessor-successor relationship with another saprotrophic mushroom species (with *C. aeruginascens* as the predecessor) due to the fact that it was typically found alone in weakly decayed logs. Other species may be found succeeding it at a deeper decay stage.

According to my results, *Xylaria* seems to prefer to fruit and feed on small and weakly decayed logs, *X. hypoxylon* in particular. With this result, speculation of pioneer species could take place. It has been found in Illinois, USA that *Xylaria* prefers to fruit during the summer season, and the month of June in particular (Hustad et al., 2011). In a different observational study, this genus was found to fruit gregariously during the months of June and July (Hipp, 2020), the beginning months of a season with high heat and humidity. With this evidence along with my study having taken place in the beginning of the rainy season in Costa Rica, this could suggest that *Xylaria* produce fruiting bodies most commonly on consistently wet logs in high heat environments. There was no mention of rich growth of *Xylaria* in August, another humid summer month, and were mainly found on weakly decayed, but also recently dead logs in this study. This combination of data is evidential support for the idea of this genus being a pioneer species. Out of the five *Xylaria* species found in the sample, only one was found on an almost decomposed log. This was specifically *X. hypoxylon*, the most abundant species in the entire sample. If *Xylaria* did not contain pioneer species, more evidence would show their presence on further decayed logs.

However, if fungal community data is taken into consideration for the specific species, *X. hypoxylon*, intermediate species status, if any at all, is supported much more than pioneer species status. This is because this species was found on six different logs which contained one or more other species of mushroom. It is much more difficult to determine succession for this species with this data. More and longer research would need to be conducted in order to determine succession with this species.

According to my results, *Ductifera pululahuana* seems to prefer to fruit and feed on medium to large logs that are weak or medium decayed. One study supports this result, finding that this genus has been most commonly found on rotting, or very decayed wood (Wells, 1958), as opposed to leaf litter which is another food source and habitat for saprotrophs. Another study further supports this location preference, particularly for this specific species, *D. pululahuana*, being found only on weathered wood or decaying hardwood trunks during the time of the study (Pratama, 2023). With this current data, succession can not be determined in this species or genus. However, with the decay data from this study, speculation may occur for *Ductifera pululahuana* to be an intermediate species, if not pioneer. Either way, further research will need to be done in order to determine its place in fungal succession if at all.

Overall, succession cannot be determined in all species of saprotrophic fungi, but suggested in a few. Genera *Chlorociboria* and *Xylaria* (excluding *X. hypoxylon*) have evidence as being pioneer species, and cannot sufficiently suggest *Ductifera* as an intermediate species. For all three genera and their included species, more research will be required to confidently determine the roles that they play in fungal succession, if at all. Specifically, a longer richness and diversity study of saprotrophic fungi would be best. This length of time for study should be longer than two weeks and potentially up to three years in order to observe during each season, multiple times. This is less so for retrieving more log samples, although important, and more for the purpose of being able to observe changes over significant time. This would yield the most accurate and specific results to prove or fail to support the idea of saprotrophic fungal succession. To particularly understand the *X. hypoxylon* role in succession, a similar long term study would be required, with an emphasis on returning to the same logs to observe which species arrive first, second, etc. Each of these future studies should also be sure to collect a much higher sample size, and with much more even sample sizes within each size and decay category.

ACKNOWLEDGEMENTS

I would like to give a special thank you to Naomi Guindon for guiding and supporting me as my primary advisor. Her help in mushroom identification and guidance was invaluable to me and my independent research project. I would like to thank my secondary advisor, Federico Chinchilla, for his support and guidance throughout project time as well. A special thank you to Frank Joyce and the Monteverde Institute for giving me the opportunity to conduct this research and explore my passions for nature. Finally, I would like to thank my homestay family for their endless support, kindness, and delicious food during the project period, as well as Alan Rockwell and Ryan Mcdowell, who helped with identification of my unknown mushroom species on iNaturalist.

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APPENDIX

Table 1. Where each species population of mushroom were found present based on log size. The small logs had the largest number of populations present and species found. The medium had the next highest number of both values, then the large logs.

Species	**Small Log**	**Medium Log**	**Large Log**
Auricularia auricularia-judae	2	-	-
Auricularia delicata	-	1	-
Auricularia mesenterica	-	-	1
Auriculariaceae Sp. 1	-	1	-
Auriculariaceae Sp. 2	1	-	-
Auriculariales Sp.	1	-	-
Chlorocibora aeruginascens	4	2	-
Clavulina coralloides	-	-	1
Clitopilus hobsonii	2	-	-
Collybiopsis neotropica	2	-	-
Cortitiaceae Sp. 1	1	-	-
Cortitiaceae Sp. 2	-	1	-
Cortitiaceae Sp. 3	-	-	1
Cortitiaceae Sp. 4	-	-	1
Cortitiaceae Sp. 5	1	-	-
Cortitiaceae Sp. 6	-	-	1
Crepidotus mollis	-	1	-
Dacryopinax martinii	1	-	-
Darcymyces stillatus	1	-	-
Ductifera pululahuana	1	3	2
Fomitopsis lilacinogilva	-	-	1
Ganoderma applanatum	-	1	-
Ganoderma brownii	1	-	-
Ganoderma Sp.	1	-	-
Heimiomyces tenuipes	-	1	-

Hemimycena lactea	-	2	-
Hymenochaete microcyla	1	-	-
Kuehneromyces mutabilis	-	-	1
Lactocollybia aurantiaca	-	-	1
Lepiota atrodisca	1	-	-
Letinus Sp.	-	-	1
Marismus epiphyllus	-	1	-
Merulius tremellosus	-	1	-
Mycena margarita	-	-	1
Orbilia xanthostigma	1	1	-
Pholiota aurivella	-	1	-
Pholiota granulosa	-	1	-
Pholiota Sp.	-	-	2
Phylacia mexicana	1	-	-
Pleurotus cornucopiae	-	1	-
Polyporales Sp.	-	1	-
Pseudohydnum gelatinosum	1	-	-
Psilocybe yungensis	-	2	-
Resupinatus merulioides	1	-	2
Rigidoporus microporus	1	-	1
Schizophyllum Sp. 1	2	-	-
Schizophyllum Sp. 2	1	-	-
Stereum versicolor	1	-	-
Stromatographium Sp.	-	-	1
Tremetes Sp.	-	-	1
Trichaptum abietinum	-	1	1
Xylaria anisopleura	-	1	-
Xylaria flabelliformis	1	-	-
Xylaria hypoxylon	4	1	2
Xylaria Sp.	-	-	1
Xylaria telfairii	1	-	-
Number of species found	26	20	19
Total number of individual populations	36	25	23

Table 2. Table of where each species population of mushroom were found present based on decay stage. The weakly decayed stage yielded the largest number of species (28) and individual populations found (36). The medium decay had the next highest species richness (12), then the recently dead tree stage (11), very decayed stage (9), and almost decomposed stage below (5). The total number of individual populations, from high to low, go in the same order as species richness.

Species	**Recently Dead Tree**	**Weakly Decayed**	**Medium Decayed**	**Very Decayed**	**Almost Decomposed**
Auricularia auricularia-judae	2	-	-	-	-
Auricularia delicata	-	1	-	-	-
Auricularia mesenterica	-	-	-	-	1
Auriculariaceae Sp. 1	1	-	-	-	-
Auriculariaceae Sp. 2	-	1	-	-	-
Auriculariales Sp.	-	1	-	-	-
Chlorocibora aeruginascens	2	4	-	-	-
Clavulina coralloides	-	-	-	1	-
Clitopilus hobsonii	-	1	1	-	-
Collybiopsis neotropica	1	1	-	-	-
Cortitiaceae Sp. 1	1	-	-	-	-
Cortitiaceae Sp. 2	-	1	-	-	-
Cortitiaceae Sp. 3	-	-	1	-	-
Cortitiaceae Sp. 4	-	1	-	-	-
Cortitiaceae Sp. 5	1	-	-	-	-
Cortitiaceae Sp. 6	-	-	-	-	1
Crepidotus mollis	-	1	-	-	-
Dacryopinax martinii	-	1	-	-	-
Darcymyces stillatus	-	1	-	-	-
Ductifera pululahuana	-	3	2	1	-
Fomitopsis lilacinogilva	-	-	1	-	-
Ganoderma applanatum	-	-	1	-	-
Ganoderma brownii	-	1	-	-	-

Ganoderma Sp.	-	1	-	-	-
Heimiomyces tenuipes	-	-	-	1	-
Hemimycena lactea	-	-	1	-	-
Hymenochaete microcyla	-	1	-	-	-
Kuehneromyces mutabilis	-	-	-	-	1
Lactocollybia aurantiaca	-	-	-	1	-
Lepiota atrodisca	-	1	-	-	-
Letinus Sp.	-	-	-	-	1
Marismus epiphyllus	-	1	-	-	-
Merulius tremellosus	-	1	-	-	-
Mycena margarita	-	1	-	-	-
Orbilia xanthostigma	-	1	-	-	-
Pholiota aurivella	-	-	-	-	-
Pholiota granulosa	1	-	-	1	-
Pholiota Sp.	-	1	1	-	-
Phylacia mexicana	-	1	-	-	-
Pleurotus cornucopiae	-	1	-	-	-
Polyporales Sp.	-	-	-	-	-
Pseudohydnum gelatinosum	1	-	-	-	-
Psilocybe yungensis	-	-	2	-	-
Resupinatus merulioides	-	-	2	1	-
Rigidoporus microporus	-	2	-	-	-
Schizophyllum Sp. 1	-	-	2	-	-
Schizophyllum Sp. 2	-	-	-	1	-
Stereum versicolor	1	-	-	-	-
Stromatographium Sp.	-	-	-	1	-
Tremetes Sp.	-	-	1	-	-
Trichaptum abietinum	-	1	-	-	-
Xylaria anisopleura	1	-	-	-	-
Xylaria flabelliformis	-	1	-	-	-
Xylaria hypoxylon	1	3	1	-	2
Xylaria Sp.	-	-	-	1	-
Xylaria telfairii	-	1	-	-	-

Number of species found	11	28	12	9	5
Total number of individual populations	13	36	16	9	6

Table 3. Which species were found together on the same logs in a community. 25 logs had more than one present fruiting body, all of these logs and their fungi are included. Fungi only seen by themselves are not included.

	Log Number																								
Species	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**6**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**
Auricularia auricularia-judae		x																							
Auricularia delicata																x									
Auricularia mesenterica	x																								
Auriculariaceae Sp. 1																	x								
Auriculariaceae Sp. 2																									x
Auriculariales Sp.													x												
Chlorocibora aeruginascens					x																				
Clitopilus hobsonii												x													
Cortitiaceae Sp. 1		x																							
Cortitiaceae Sp. 2					x																				
Cortitiaceae Sp. 3								x																	

Cortitiaceae Sp. 4														x											
Cortitiaceae Sp. 5																		x							
Cortitiaceae Sp. 6																								x	
Dacryopinax martinii																									x
Darcymyces stillatus				x																					
Ductifera pululahuana				x	x				x										x		x				
Fomitopsis lilacinogilva																					x				
Ganoderma applanatum																						x			
Ganoderma brownii																				x					
Heimiomyces tenuipes											x														
Hemimycena lactea															x										
Hymenochaete microcyla										x															
Kuehneromyces mutabilis																								x	
Lactocollybia aurantiaca							x																		
Lepiota atrodisca																				x			x		
Letinus Sp.			x																						
Marismus epiphyllus																			x						
Psilocybe yungensis																						x	x		
Orbilia xanthostigma												x				x									
Pholiota aurivella																	x								

Pholiota granulosa										x															
Phylacia mexicana									x																
Pleurotus cornucopiae						x																			
Polyporales Sp.															x										
Resupinatus merulioides							x														x				
Rigidoporus microporus													x												
Stromatographium Sp.									x																
Tremetes Sp.								x																	
Trichaptum abietinum														x		x									
Xylaria anisopleura																	x								
Xylaria hypoxylon	x		x			x							x					x							x
Xylaria Sp.									x																
Xylaria telfairii													x												